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United States Patent	12416156
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Xing; Linlin et al.

Colored roofing membrane with improved solar reflectance

Abstract

A multilayer roofing material includes a first layer, a second layer, a third layer, and a fourth layer. The first layer is a topmost layer and a color layer. The first layer comprises a UV stabilizer. The second layer comprises TPO. The second layer is a white layer. The third layer comprises a scrim. The third layer contacts the second layer and the second layer is between the first layer and third layer. The fourth layer comprises TPO. The fourth layer is a core layer. The fourth layer contacts the third layer.

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Appl. No.:	19/202135
Filed:	May 08, 2025

Prior Publication Data

Document Identifier	Publication Date
US 20250263928 A1	Aug. 21, 2025

Related U.S. Application Data

continuation parent-doc US 18789208 20240730 PENDING child-doc US 19202135
continuation parent-doc US 17820936 20220819 US 12077965 20240903 child-doc US 18789208
continuation parent-doc US 16601151 20191014 US 11421419 20220823 child-doc US 17820936
continuation parent-doc US 14213299 20140314 US 10480192 20191119 child-doc US 16601151
us-provisional-application US 61793758 20130315

Publication Classification

Int. Cl.: E04D5/10 (20060101); E04D5/06 (20060101)

U.S. Cl.:

CPC E04D5/10 (20130101); E04D5/06 (20130101); Y02A30/254 (20180101); Y02B80/00 (20130101); Y10T156/10 (20150115); Y10T428/31913 (20150401); Y10T442/10 (20150401)

Field of Classification Search

CPC: E04D (5/10); E04D (5/06); Y02A (30/254); Y02B (80/00); Y10T (156/10); Y10T (428/31913); Y10T (442/10)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3358355	12/1966	Youssi et al.	N/A	N/A
4386981	12/1982	Clapperton	N/A	N/A
4584220	12/1985	Melbye	N/A	N/A
4992315	12/1990	Zickell et al.	N/A	N/A
6415574	12/2001	Beck	N/A	N/A
7070843	12/2005	Bartek et al.	N/A	N/A
7604707	12/2008	Bartek et al.	N/A	N/A
7772136	12/2009	Arthurs et al.	N/A	N/A
7776417	12/2009	Mohseen et al.	N/A	N/A
7882671	12/2010	Douglas et al.	N/A	N/A
8833037	12/2013	French et al.	N/A	N/A
9163410	12/2014	French et al.	N/A	N/A
9751286	12/2016	Yoda et al.	N/A	N/A
9920515	12/2017	Xing et al.	N/A	N/A
10081945	12/2017	Peng et al.	N/A	N/A
2004/0161569	12/2003	Zanchetta et al.	N/A	N/A
2007/0077838	12/2006	Binkley et al.	N/A	N/A
2010/0178827	12/2009	Thai et al.	N/A	N/A
2010/0255739	12/2009	Black et al.	N/A	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 18/789,208, filed on Jul. 30, 2024, which is a continuation of U.S. patent application Ser. No. 17/820,936, filed on Aug. 19, 2022, now U.S. Pat. No. 12,077,965,

which is a continuation of U.S. patent application Ser. No. 16/601,151, filed on Oct. 14, 2019, now U.S. Pat. No. 11,421,419, which is a continuation of U.S. patent application Ser. No. 14/213,299, filed on Mar. 14, 2014, now U.S. Pat. No. 10,480,192, which claims priority to U.S. Provisional Application Ser. No. 61/793,758, filed Mar. 15, 2013, the entire disclosures of which are herein incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates to exemplary embodiments of a thermoplastic polyolefin roofing membrane, and more particularly, to exemplary embodiments of a thermoplastic polyolefin roofing membrane with an improved solar reflectance.

BACKGROUND INFORMATION

(2) Conventional thermoplastic polyolefin (“TPO”) based roofing membranes can be made up of three layers. As shown in FIG. 1, a typical TPO based roofing membrane **100** can have an upper layer **110** that is generally a TPO sheet, sometimes referred to as a “cap layer”, which can be formulated with ultraviolet absorbers and stabilizers, thermal stabilizers, pigments, fire retardants and the like. Typically, the upper layer **110** can be formulated to provide for long term stability in an outdoor environment. The bottom layer **130** of a TPO roofing membrane can also be a TPO sheet, sometimes referred to as a “core layer”, and generally can contain fewer or lower concentrations of stabilizers.

(3) The central layer **120** of a TPO roofing membrane can typically be a fabric scrim that acts as a reinforcement to improve the physical performance of the roofing membrane **100**. The fabric scrim **120** can contribute significantly to the tensile strength of the roofing membrane **100** and provide for dimensional stability. Typically, the fabric reinforcement can be a polyester based scrim. Glass based scrims can also be used where additional weight or improved dimensional stability may be required.

(4) The three layers can be laminated together at elevated temperatures such that the upper layer **110** and bottom layer **130** can be joined or fused together through the interstices of the scrim of the middle layer **120**.

(5) Roofing membranes can provide more benefits than just protection from the weather. In addition to aesthetic aspects of residential roofing material design, both residential and commercial roofing can now provide advantages to the energy usage of buildings and homes. By providing roofing membranes with higher reflectance, it can be possible to lower the heat gain experienced by a building. Such reflective systems are referred to as “cool roofs” and provide for lowered cooling demand for building interiors.

(6) TPO roofing membranes are mostly used on low slope roofs that can have pitches of less than approximately two inches vertical rise over an approximately twelve inch horizontal distance. One advantage of TPO membranes compared to traditional asphaltic systems can be their high total solar reflectance (“TSR”), which can be achieved through various means including the incorporation of titanium dioxide.

(7) The TSR can indicate how much of the sun's ultraviolet, visible and infrared energy (i.e., solar flux) is reflected. TSR can be defined as the fraction of solar flux reflected by a surface expressed as a percent or within the range of 0.00 and 1.00. Measurement can be done using the methods described by the American Society for Testing and Materials (“ASTM”) Standard E903, ASTM C1549 or ASTM E1918. ASTM E903 measures TSR over the wavelength range 250 nm to 2500 nm using integrating spheres. By using a commercial portable solar reflectometer that can be calibrated using specimens of known TSR, ASTM C1549 can determine TSR from measurements at four wavelengths in the solar spectrum: 380 nm, 500 nm, 650 nm and 1220 nm. ASTM C1549 shows the comparison results between the C1549 and E903 methods. The solar reflectance results at air mass 1.5 measured per ASTM C1549 are generally 1.9% greater than those obtained with ASTM E903. ASTM discontinued E903 in August 2005 in accordance with section 10.5.3.1 of the

Regulations Governing ASTM Technical Committees, which requires that standards shall be updated by the end of the 8^{sup}.th year since the last approval date. ASTM E1918 covers the TSR measurement for various horizontal and low sloped surfaces and materials in the field, using a pyranometer. This test method can be used when the sun angle to the normal from a surface is less than 45 degrees. For commercially available white or very light color TPO membrane, TSR is typically 0.70 or higher.

(8) Roofs that can be seen from the ground and roofs that are overlooked by taller buildings are generally unattractive when white and highly reflective. Therefore, manufacturers have begun to use color pigments for those TPO roofing membranes that will be viewed, or that are required to have aesthetic value. However, such colored membranes have significantly degraded TSR values, compared to white, depending on the darkness of the pigment used.

(9) Some TPO roofing membranes have a “cool color” pigment technology that reflects more infrared radiation than would be expected for a given color. However, these roofing membranes provide only a marginal improvement over standard colored TPO roofing membranes.

(10) The cap layer, or upper layer, of a roofing membrane represents a very significant portion of the cost of a roofing membrane. Further, in hot climates, the solar reflectance of colored TPO membranes can affect the heat gain experienced by a commercial or residential building. Also, the long term weathering performance and color fastness of the cap layer of colored TPO roofing membranes are critical for long term waterproofing and for aesthetic appearance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other exemplary objects of the present disclosure will be apparent upon consideration of the following detailed description, taken in conjunction with the accompanying exemplary drawings and claims, in which like reference characters refer to like parts throughout, and in which:

(2) FIG. 1 is an illustration of a conventional thermoplastic polyolefin based roofing membrane;

(3) FIG. 2 is an illustration of an exemplary embodiment of a thermoplastic polyolefin roofing membrane according to the present disclosure; and

(4) FIG. 3 is an illustration of another exemplary embodiment of a thermoplastic polyolefin roofing membrane according to the present disclosure.

(5) Throughout the figures, the same reference numerals and characters, unless otherwise stated, are used to denote like features, elements, components or portions of the illustrated embodiments. Moreover, while the subject invention will now be described in detail with reference to the figures, it is done so in connection with the illustrative embodiments. It is intended that changes and modifications can be made to the described embodiments without departing from the true scope and spirit of the subject disclosure.

SUMMARY OF EXEMPLARY EMBODIMENTS OF THE DISCLOSURE

(6) At least some of the above described problems can be addressed by exemplary embodiments of the colored thermoplastic polyolefin roofing membrane having an enhanced solar reflectance according to the present disclosure.

(7) In some embodiments, a multilayer roofing material includes a first layer, wherein the first layer is a topmost layer, wherein the first layer is a color layer, wherein the first layer comprises a UV stabilizer; a second layer, wherein the second layer comprises TPO, wherein the second layer is a white layer; a third layer, wherein the third layer comprises a scrim, wherein the third layer contacts the second layer, wherein the second layer is between the first layer and third layer; and a fourth layer, wherein the fourth layer comprises TPO, wherein the fourth layer is a core layer, and wherein the fourth layer contacts the third layer.

- (8) In some embodiments, the scrim is a reinforcing polyester scrim. In some embodiments, the first layer comprises a color pigment. In some embodiments, the second layer comprises at least one of a titanium dioxide pigment, a UV stabilizer package, or a fire retardant.
- (9) In some embodiments, the fourth layer has a dark color. In some embodiments, the fourth layer comprises at least one of carbon black, titanium dioxide, a UV stabilizer package, or a fire retardant.
- (10) In some embodiments, the first layer comprises at least two sublayers. In some embodiments, the second layer comprises at least two sublayers.
- (11) In some embodiments, the fourth layer is a dark color and comprises at least one of carbon black, titanium dioxide, a UV stabilizer package, or a fire retardant.
- (12) In some embodiments, the first layer is a thin layer. In some embodiments, In some embodiments, the fourth layer is about 22 mil. In some embodiments, at least one of the first, second, third or fourth layers comprises a film.
- (13) In some embodiments, the second layer has a very high solar reflectance. In some embodiments, the first layer has a high solar reflectance. In some embodiments, the solar reflectance is measured in accordance with the standards of ASTM C1549 test standards.
- (14) In some embodiments, a multilayer roofing material includes a first layer, wherein the first layer is a topmost layer, wherein the first layer is a color layer, wherein the first layer comprises a thermal stabilizer; a second layer, wherein the second layer comprises TPO, wherein the second layer is a white layer; a third layer, wherein the third layer comprises a scrim, wherein the third layer contacts the second layer, wherein the second layer is between the first layer and third layer; and a fourth layer, wherein the fourth layer comprises TPO, wherein the fourth layer is a core layer, and wherein the fourth layer contacts the third layer.
- (15) In some embodiments, the scrim is a reinforcing polyester scrim.
- (16) In some embodiments, the first layer comprises a color pigment. In some embodiments, the second layer comprises at least one of a titanium dioxide pigment, or a fire retardant. In some embodiments, the fourth layer has a dark color.
- (17) In some embodiments, the fourth layer comprises at least one of a UV stabilizer package, or a fire retardant.
- (18) In some embodiments, the first layer comprises at least two sublayers. In some embodiments, the second layer comprises at least two sublayers.
- (19) In some embodiments, the fourth layer is a dark color and comprises at least one of carbon black, titanium dioxide or a fire retardant.
- (20) In some embodiments, the first layer is a thin layer. In some embodiments, the fourth layer is about 22 mil.
- (21) In some embodiments, at least one of the first, second, third or fourth layers comprises a film.
- (22) In some embodiments, the second layer has a very high solar reflectance. In some embodiments, the first layer has a high solar reflectance. In some embodiments, the solar reflectance is measured in accordance with the standards of ASTM C1549 test standards.
- (23) In some exemplary embodiments of the present disclosure, a thermoplastic polyolefin membrane is provided, comprising a cap layer comprising a top color layer, and a white sub-layer having a solar reflectance underneath the top color layer, and a core layer underneath the cap layer. The thermoplastic polyolefin membrane can further comprise a fabric scrim between the cap layer and the core layer. The fabric scrim can be a reinforcing polyester scrim.
- (24) The top color layer can comprise one or more of cool color pigments, UV stabilizers, and fire retardants. The white sub-layer can comprise one or more of titanium dioxide pigment, UV stabilizer package, fire retardant and TPO resins. The white sub-layer can comprise one or more layers. The core layer can comprise a dark color, and can comprise carbon black, titanium dioxide, UV stabilizer packages, fire retardant and TPO resins. The core layer can comprise a light color, and can comprise titanium dioxide, UV stabilizer packages, fire retardant and TPO resins.

(25) In some exemplary embodiments of the present disclosure, a method of manufacturing a thermoplastic polyolefin membrane can be provided, the method comprising providing a cap layer comprising providing a top color layer, and providing a white sub-layer having a solar reflectance underneath the top color layer, and providing a core layer underneath the cap layer. The method can further comprise providing a fabric scrim provided between the cap layer and the core layer. The fabric scrim can be a reinforcing polyester scrim.

(26) These and other objects, features and advantages of the present disclosure will become apparent upon reading the following detailed description of embodiments of the present disclosure, when taken in conjunction with the appended claims.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS OF DISCLOSURE

(27) Exemplary embodiments of the colored thermoplastic polyolefin roofing membrane with an improved solar reflectance according to the present disclosure will be described below.

(28) FIG. 2 is an illustration of an exemplary embodiment of a thermoplastic polyolefin roofing membrane **200** according to the present disclosure. A cap layer (i.e., a top layer) can be provided having a thin top color layer **210** and a white sub-layer **220** provided underneath the thin top color layer **210**. The TPO roofing membrane **200** can optionally be provided with a fabric scrim, such as a reinforcing polyester scrim **230**, which can be provided underneath the one or more white sub-layers **220**. The white sub-layer **220** can be made up of one or more layers. The white sub-layer can comprise one or more of titanium dioxide pigment, standard UV stabilizers, fire retardant and TPO resins. A black or a dark color can be provided as a core layer **240**. The dark core layer **240** can comprise a combination of carbon black, titanium dioxide, UV stabilizers, fire retardant and/or TPO resins.

(29) FIG. 3 is an illustration of another exemplary embodiment of a thermoplastic polyolefin roofing membrane **300** according to the present disclosure. A cap layer (i.e., a top layer) can be provided having a thin top color layer **310** and a white sub-layer **320** provided underneath the thin top color layer **310**. The TPO roofing membrane **300** can optionally be provided with a reinforcing polyester scrim **330**, which can be provided underneath the one or more white sub-layers **320**. The white sub-layer **320** can be made up of one or more layers. The white sub-layer can comprise one or more of titanium dioxide pigment, standard UV stabilizers, fire retardant and TPO resins. A white or light cool color(s) can be provided as a core layer **340**. The white core layer **340** can comprise one or more of titanium dioxide, UV stabilizers, fire retardant and TPO resins.

(30) In the exemplary embodiments of the present disclosure, the top color TPO layer can contain one or more of weatherable cool color pigments, UV stabilizers, and fire retardants for weathering, color fastness, fire resistance and energy cost savings due to the cool colors with a high solar reflectance. The white sub-layer underneath the thin top color layer can be white with very high solar reflectance to further enhance the solar reflectance of the top color TPO layer. Different UV stabilizer packages from the top color TPO layer can be used in the white sub-layer(s).

(31) In another exemplary embodiment of the present disclosure, a high strength white polypropylene film can be provided. The high strength white polypropylene film can be produced by, for example, a blown film process. The high strength white polypropylene film can replace any of the following components in the TPO membrane construction: the sub-white layer in the cap layer, the reinforcing polyester scrim, or any colored core layer.

(32) In an exemplary embodiment of the present disclosure, as provided below, a higher TSR Regal Red TPO Membrane can be provided.

Example-1. Higher TSR Regal Red TPO Membrane

(33) TABLE-US-00001 % improvement over Comparative Example -1 Example 1
Example 1 Cap Layer Construction Top color layer: 10 mil Regal Red Single color layer: 23 TPO mil Regal Red TPO Sub-layer: 13 mil white TPO Cap Layer Compositions Top color layer: same formulation as Regal red color that of Example 2 (comparative) pigments, excellent Sub-layer: TiO₂.sub.2 pigment, standard UV stabilizer UV stabilizer package, fire retardant, package, fire TPO

resins retardant, TPO resins Reinforced Scrim Polyester Polyester Core Layer Construction & 22 mil dark gray TPO 22 mil dark gray TPO Compositions Same formulation as that of Carbon black, TiO.sub.2, Comparative Example 1 fire retardant, UV stabilizer package, TPO resins Total Solar Reflectance 0.48 0.32 48% (TSR) per ASTM C1549 Color Fastness, Delta E 1.6 3.6 56% (after 8820 kJ/m.sup.2 exposure in Xenon Arc WOM per ASTM D6878)

(34) In the chart above, Example 1 describes a TPO roofing membrane according to the exemplary embodiments of the present disclosure, and Comparative Example 1 provides a description of a known roofing membrane. In Example 1, a top color layer comprising 10 mil Regal Red is provided with a white sub-layer comprising 13 mil white TPO. The white sub-layer comprises titanium dioxide pigment, standard UV stabilizer package(s), fire retardant and TPO resins. The Comparative Example 1 provides a top layer having a single color layer having 23 mil Regal Red TPO. The top layer of Comparative Example 1 has a similar composition to Example 1. A similar reinforced scrim and core layer construction and composition is provided in Example 1 and Comparative Example 1. As shown in the chart, the total solar reflectance of Example 1 provided a 48% improvement compared to Comparative Example 1, and a color fastness providing a 56% improvement over Comparative Example 1.

(35) In another exemplary embodiment of the present disclosure, as provided below, a higher TSR Goldenrod TPO Membrane can be provided.

Example-2. Higher TSR Goldenrod TPO Membrane

(36) TABLE-US-00002 % improvement over Comparative Comparative Example-2 Example 2
Example 2 Cap Layer Construction Top color layer: 10 mil Goldenrod Single color layer: TPO 23 mil Goldenrod Sub-layer: 13 mil white TPO TPO Cap Layer Compositions Top color layer: same formulation as Goldenrod color that of Example 2 (comparative) pigments, excellent Sub-layer: TiO.sub.2 pigment, standard UV UV stabilizer stabilizer package, fire retardant, TPO package, fire resins retardant, TPO resins Reinforced Scrim Polyester Polyester Core Layer Construction & 22 mil dark gray TPO 22 mil dark gray Compositions Same formulation as that of TPO Comparative Example 1 Carbon black, TiO.sub.2, fire retardant, UV stabilizer package, TPO resins Total Solar Reflectance 0.64 0.56 14% (TSR) per ASTM C1549 Color Fastness, Delta E 0.7 4.1 83% (after 6800 kJ/m² exposure in Xenon Arc WOM per ASTM D6878)

(37) In the chart above, Example 2 describes a TPO roofing membrane according to the exemplary embodiments of the present disclosure, and Comparative Example 2 provides a description of a known roofing membrane. In Example 2, a top color layer comprising 10 mil Goldenrod TPO is provided with a white sub-layer comprising 13 mil white TPO. The white sub-layer comprises titanium dioxide pigment, standard UV stabilizer package(s), fire retardant and TPO resins. The Comparative Example 2 provides a top layer having a single color layer having 23 mil Goldenrod TPO. The top layer of Comparative Example 2 has a similar composition to Example 2. A similar reinforced scrim and core layer construction and composition is provided in Example 2 and Comparative Example 2. As shown in the chart, the total solar reflectance of Example 2 provided a 14% improvement compared to Comparative Example 2, and a color fastness providing a 83% improvement over Comparative Example 2.

(38) Various other considerations can also be addressed in the exemplary embodiments of the colored TPO roofing membrane of the present disclosure. For example, different configurations of layers and sub-layers can be provided to provide a high solar reflectance, including various combinations of cool colors and white layers.

(39) Further, there are many advantages that can be provided by the exemplary embodiments according to the present disclosure. For example, with the reduced heat absorption provided by high solar reflectance cool colored roofing materials, not only are there environmental benefits, such as energy saving and reducing heat island effect, but there are also potential improvements to the longevity of roofing materials. Since heat accelerates the degradation of polymeric membranes, technology that can lower membrane temperatures can result in better retention of their physical

properties. The reduced thermal expansion and contraction due to the reduced temperatures with high solar reflectance also helps to extend the life of the roofing materials.

(40) The foregoing merely illustrates the principles of the disclosure. Various modifications and alterations to the described embodiments will be apparent to those skilled in the art in view of the teachings herein. It will thus be appreciated that those skilled in the art will be able to devise numerous layers, configurations, arrangements, manufacture and methods which, although not explicitly shown or described herein, embody the principles of the disclosure and are thus within the spirit and scope of the disclosure.

Claims

1. A multilayer roofing material, comprising: a first layer, wherein the first layer is a topmost layer, wherein the first layer is a color layer, wherein the first layer comprises a UV stabilizer; a second layer, wherein the second layer comprises TPO, wherein the second layer is a white layer; a third layer, wherein the third layer comprises a scrim, wherein the third layer contacts the second layer, wherein the second layer is between the first layer and third layer; and a fourth layer, wherein the fourth layer comprises TPO, wherein the fourth layer is a core layer, wherein the fourth layer contacts the third layer.
2. The multilayer roofing material of claim 1, wherein the scrim is a reinforcing polyester scrim.
3. The multilayer roofing material of claim 1, wherein the first layer comprises a color pigment.
4. The multilayer roofing material of claim 1, wherein the second layer comprises at least one of a titanium dioxide pigment, a UV stabilizer package, or a fire retardant.
5. The multilayer roofing material of claim 1, wherein the fourth layer has a dark color.
6. The multilayer roofing material of claim 1, wherein the fourth layer comprises at least one of carbon black, titanium dioxide, a UV stabilizer package, or a fire retardant.
7. The multilayer roofing material of claim 1, wherein the first layer comprises at least two sublayers.
8. The multilayer roofing material of claim 1, wherein the second layer comprises at least two sublayers.
9. The multilayer roofing material of claim 1, wherein the fourth layer is a dark color and comprises at least one of carbon black, titanium dioxide, a UV stabilizer package, or a fire retardant.
10. The multilayer roofing material of claim 1, wherein the fourth layer is about 22 mil.
11. The multilayer roofing material of claim 1, wherein at least one of the first, second, third or fourth layers comprises a film.
12. A multilayer roofing material, comprising: a first layer, wherein the first layer is a topmost layer, wherein the first layer is a color layer, wherein the first layer comprises a thermal stabilizer; a second layer, wherein the second layer comprises TPO, wherein the second layer is a white layer; a third layer, wherein the third layer comprises a scrim, wherein the third layer contacts the second layer, wherein the second layer is between the first layer and third layer; and a fourth layer, wherein the fourth layer comprises TPO, wherein the fourth layer is a core layer, wherein the fourth layer contacts the third layer.
13. The multilayer roofing material of claim 12, wherein the scrim is a reinforcing polyester scrim.
14. The multilayer roofing material of claim 12, wherein the first layer comprises a color pigment.
15. The multilayer roofing material of claim 12, wherein the second layer comprises at least one of a titanium dioxide pigment, or a fire retardant.
16. The multilayer roofing material of claim 12, wherein the fourth layer has a dark color.
17. The multilayer roofing material of claim 12, wherein the fourth layer comprises at least one of a UV stabilizer package, or a fire retardant.
18. The multilayer roofing material of claim 12, wherein the first layer comprises at least two

sublayers.

19. The multilayer roofing material of claim 12, wherein the second layer comprises at least two sublayers.

20. The multilayer roofing material of claim 12, wherein the fourth layer is a dark color and comprises at least one of carbon black, titanium dioxide or a fire retardant.

21. The multilayer roofing material of claim 12, wherein the fourth layer is about 22 mil.

22. The multilayer roofing material of claim 12, wherein at least one of the first, second, third or fourth layers comprises a film.
